

Investigating Lighting Quality in Office Workstations: A Combined Approach Utilizing Virtual Reality and Physical Workstations

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ABSTRACT

While recent studies suggest benefits of using immersive virtual reality (VR) technology to obtain user experience during the design process of spaces, it is still unclear whether VR can be used as a reliable tool to evaluate certain environmental features such as lighting quality. This study examines six daylight control options (electrochromic windows versus blinds with different degrees of daylight transmission and color) and their impacts on users' satisfaction and visual comfort in a physical office as well as its digital twin. The daylight control options were installed in a physical office setting located in Mississippi, U.S., where participants (N = 112) experienced each daylight control option and completed an online survey. Using Matterport, web-based 3D models of the office with the daylight control options were created. Participants (N = 35, study still in progress) explored the immersive virtual models using Oculus Quest 2 VR goggles on a university campus. A verbal survey was administered to obtain participants' feedback while they were experiencing the virtual office using VR. Semi-structured interviews were also conducted after participants' VR experience. Findings revealed high visual comfort ratings for electrochromic windows in both virtual and physical settings. However, results regarding perception of glare severity and overall satisfaction with workstations were inconsistent between the VR and physical setting. Findings from this study help the designers, architects, and stakeholders identify the optimal window design features to improve user experience and comfort for office workstations. Also, this study addresses the validity of VR as a tool to assess user experience of lighting quality in spaces.

Keywords: daylight control, virtual reality, visual comfort, satisfaction.

1 INTRODUCTION

Based on the existing body of research, lighting has a significant impact on shaping the user's experience and perception of the space. In terms of working environments, studies delineated the prominent effect of lighting in individual's health and performance in such environments [1][2][3]. While daylight has been identified as a superior source of lighting due to its availability and wide spectrum to enhance workers' performance

and health in office settings [4], there is a paucity of research regarding improving the quality of interior lighting as it relates to comfort and satisfaction of office workers using daylight control systems [5]. To date, studies have used different methodologies to investigate user experience in such settings, including immersive VR tools [6][7]. With the current technology, very high-resolution photos of the physical environments are used in VR to create the clone of the physical environment. While some studies show a high level of perceptual accuracy, it is still unclear whether VR is a reliable tool for obtaining user feedback regarding certain environmental features such as lighting quality in office spaces as a standalone tool [8][9].

This study examines six daylight control options (electrochromic windows versus blinds with different degrees of daylight transmittance) and their impacts on users' satisfaction regarding lighting quality and visual comfort. The study incorporates realistic 3D modeling of the office as the study setting via Matterport instead of using architectural software to simulate the lighting conditions in the 3D space. Using data collection in the physical office setting, the study also seeks to understand validity and reliability of VR as a tool for evaluation of perceived lighting quality during the design process. The study addresses the following research questions:

1. How do different daylight control options for windows impact perceived visual comfort and satisfaction with lighting quality in office workstations?
2. How do different daylight control options for windows impact overall satisfaction with the office environment?
3. What are the similarities and differences in the perception of these factors in virtual models produced by Matterport versus the physical office setting?

2 METHODS

This IRB-approved, multi-methods study was conducted in two phases. Phase one was conducted in the physical office setting located in Mississippi, U.S., where participants experienced each daylight control option and filled in an online survey. In phase two of the study, participants were recruited on a university campus and experience the virtual model of the same office environment simulated in Matterport. The participants were familiarized with the virtual environment prior to the study. After the orientation phase, participants were immersed in VR using the Oculus Quest 2 VR Goggles (Model KW49CM). An online survey in Qualtrics platform was administered verbally after participants were randomly assigned to the workstations and experienced the daylight control options, including windows with blinds versus electrochromic windows. The survey items evaluate participants' perception of the lighting quality and visual comfort based on a five-point Likert scale (1 being low rating and 5 being a high rating) as well as the overall satisfaction level on a scale of 1 to 10 (1 being unsatisfied and 10 being satisfied). Semi-structured interviews were also used to obtain in-depth user feedback on each lighting control option after their experience of the virtual office environment.

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2.1 Analytical Procedure

The survey data were analyzed using JMP Pro (version 14.0). Participants' ratings of 4 and 5 were coded as a high rating, and ratings of 1 to 3 were coded as a neutral or low rating. Descriptive analysis was used to report response percentages for each survey item based on the recoded data. Semi-structured interviews were audio recorded, transcribed verbatim, and prepared for content analysis via MAXQDATA (2018).

2.2 The Study Setting

The office space is 30' x 40' with six full-height glazing panels (4' 8" x 9'10") facing east and south. Each glazing facing east is covered with different daylight control systems, including static dark shades with visible light transmittance (Tvis) of 1%, 4%, 6%, and 11%, and electrochromic windows (View Smart Windows) with Tvis of 1% and 0.5% (Figure 1). A workstation is placed adjacent to each window panel facing east. The workstations are separated via partitions with zero light transmittance. Using Matterport tripod-mounted 3D camera, the images of the office interior were captured after the installation of daylight control options at each workstation. The images were transferred to 3D Capture app and Matterport's cloud server for creating a realistic 3D model. The 3D models were uploaded on a web browser using the Unity multimedia plug-in to be experienced by users worldwide via the Oculus Quest VR platform linked to the web browser.

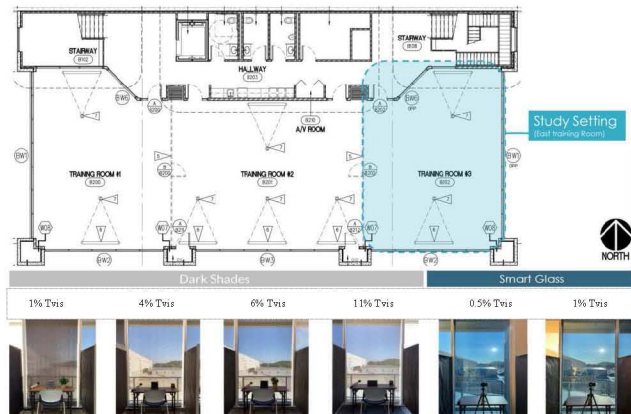


Figure 1: The office workstations with different daylight control options in the physical office setting.

2.3 The Study Procedure

In phase one, participants from the general public (N = 112) were recruited on September 27th and 28th, 2021, at the office location in Mississippi, U.S. Participants were randomly assigned to the office workstations and filled in an online survey while sitting at the workstation. In phase two, participants (N = 35, study still in progress) were recruited on a university campus located in Michigan, U.S. The majority of participants (70%) were male and aged between 18 and 29 years old. The participants were faculty members, students, and employees at a university in Michigan, U.S. who were recruited on campus through an email invitation between November 15th to December 14th, 2021. The data collection for the second phase of the study was resumed on January 20th, 2022, and is currently in progress.

3 FINDINGS

Preliminary findings pertaining to participants' subjective visual comfort indicated high ratings (ratings of 4 or 5 on a five-point Likert scale) for the electrochromic windows with Tvis of 1%. Electrochromic windows also had lower daylight glare probabilities and lower temperatures as measured by high dynamic range and infrared cameras. In the physical office, 89% of participants highly rated the electrochromic windows (Tvis = 1%) and 65% highly rated the dark shade (Tvis = 6%) for their visual comfort, which was the highest rated of the 4 shades. Ratings from the virtual office indicated that 75% of participants highly rated both the electrochromic windows (Tvis = 1%) and the dark shade (Tvis = 6%) for their visual comfort. Results related to natural lighting satisfaction in the physical office showed that the electrochromic windows with Tvis of 1% and 0.5% outperformed all dark shades, and were highly rated by 86% and 87% of participants, respectively. In the virtual office, however, the electrochromic windows (Tvis = 1%) and dark shade (Tvis = 11%) were highly rated by 100% of the participants. Ratings for the overall satisfaction with the office workstations were also inconsistent between the physical and virtual offices. The office workstation with the electrochromic windows (Tvis = 1%) received the highest rating (8.9 out of 10) in the physical office while the dark shade (Tvis = 11%) with a rating of 7.3 out of 10 was the highest rated workstation in the virtual office. This finding contradicted the fact that the 11% dark shade had the highest daylight glare probability (DGP = 0.46) in the physical office which falls in the intolerable range for occupant visual comfort.

Results from the semi-structured interviews following the VR experience revealed that glare was perceived to be a potential problem for the virtual workstation with the 11% dark shade; however, it did not have a major impact on participants' satisfaction level. Participants stated: "it was a bit glary, like the window had a bit too much glare. Other than that, it was good. It was bright enough. You could see everything in the room very clearly." or "I was concerned about the sun when working with a laptop in the workstation because of the glare!". In addition to glare, one participant mentioned thermal comfort as a potential problem stating: "I can imagine that the space may be very warm especially staying eight hours a day". On the other hand, electrochromic windows (Tvis = 1%) were appreciated by participants as they could provide natural lighting and a bright working environment. One participant stated: "this workstation was brighter and sunnier than the workstations I have already worked in". Also, two participants assigned to this workstation mentioned that they preferred to have adjustable blinds in addition to the electrochromic feature of the window. For example, participants mentioned: "I think having some kind of blinds that can be adjusted during the day when the sun is direct would be better" or "it would have been nice if the windows were operable, or if there were blinds that you could roll down to give yourself more shade".

4 DISCUSSION

The findings showed that participants highly rated the electrochromic windows for visual comfort, thermal comfort, satisfaction with natural lighting, and overall satisfaction compared to the dark shades in the physical office. However, comparing the ratings between the physical and virtual office workstations revealed differences in evaluation of the categories mentioned above, especially in the overall satisfaction ratings. This finding is in line with reports from a recent study that suggested differences in user perceptions regarding certain environmental features, including color and texture, in virtual

versus physical environments [9]. Even using the 3D imagery of the shading materials in the physical office using Matterport and immersive VR experience of the virtual model still led to differences in overall satisfaction and evaluation of natural lighting quality in the office interior, most likely because these technologies do not accurately represent the intensity of the sun and may minimize glare experiences. Moreover, the shade that was highly rated regarding participant satisfaction in VR had the highest daylight glare probability in reality, indicating that VR headsets and Matterport 3D imagery were not well suited to represent severity of glare that can negatively influence occupant experience in reality. Despite limitations associated with VR tools and Matterport 3D imagery, findings delineated consistencies in electrochromic windows ratings for natural lighting satisfaction and glare detection between the virtual and physical office.

5 CONCLUSION

This study examined different daylight control options in an office environment using a combined methodology incorporating evaluations in physical and virtual settings. Unlike conventional methods in built environment research with VR, the virtual model was created using 3D imagery via Matterport in an office setting with physical prototypes for daylight control installed on windows. The preliminary analysis of data revealed some inconsistencies in the overall satisfaction ratings between the two settings. Future research can test this methodology in other environments with a wider population using different VR headsets and equipment to further investigate the efficiency of immersive VR tools for investigating lighting quality in interior spaces.

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